

Year 7 • Chapter 8 • Area and volume: answers (teacher copy)

8.01 Units of measurement (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: 3.2 km
2. Ex 1b: 42 g
3. Ex 2a: 8300 mL
4. Ex 2b: 4 h
5. Ex 3a: 54 000 s
6. Ex 3b: 6 307 200 min (B)

Easy • Set A (quick drill)

- | | | |
|-------------|-------------|-------------|
| 1. 3.2 km | 6. 0.125 kL | 11. 180 min |
| 2. 9040 mm | 7. 5000 m | 12. 1.5 min |
| 3. 7.15 m | 8. 3.5 m | 13. 60 mm |
| 4. 42 g | 9. 2400 g | 14. 4 t |
| 5. 0.0035 t | 10. 1.5 L | |

Easy • Set B

1. 8300 mL
2. 4 200 000 mL
3. 4 h
4. 900 min
5. 60 h
6. 17 kg

Medium • Set C

1. 54 000 s
2. 65 000 cm
3. 4.5 kg
4. 200 min
5. 0.75 L
6. 10 080 min

Challenging • Set D

1. 6 307 200 min
2. 4.2 kg
3. 7.2 L
4. 950 m, 1 100 000 mm, 1.2 km, 125 000 cm

Skill drill 1 (Easy basics)

- | | | |
|---------------|---------------|-------------|
| 1. 22 cm | 15. 57 000 mg | 29. 300 min |
| 2. 5 mm | 16. 4.2 t | 30. 144 h |
| 3. 36 m | 17. 4 m | 31. 600 s |
| 4. 280 L | 18. 5 km | 32. 360 s |
| 5. 48 000 mm | 19. 27 m | 33. 11 h |
| 6. 0.13 L | 20. 250 mm | 34. 72 h |
| 7. 23 000 mm | 21. 40 g | 35. 96 h |
| 8. 0.15 m | 22. 0.43 m | 36. 3 h |
| 9. 0.45 kL | 23. 540 mL | 37. 480 s |
| 10. 53 000 mL | 24. 26 000 m | 38. 6 h |
| 11. 3 300 mg | 25. 180 s | 39. 420 s |
| 12. 31 000 m | 26. 2 h | 40. 60 min |
| 13. 0.03 kg | 27. 120 min | |
| 14. 0.04 g | 28. 420 min | |

Skill drill 2 (Medium basics)

- | | | |
|--------------|--------------|------------|
| 1. 570 cm | 10. 8.2 kg | 19. 8.75 h |
| 2. 51 000 mg | 11. 910 mm | 20. 4.25 h |
| 3. 3 000 mm | 12. 1 700 m | 21. 9.5 h |
| 4. 830 cm | 13. 9 600 cm | 22. 42 h |
| 5. 55 m | 14. 4.6 kg | 23. 162 h |
| 6. 180 cm | 15. 50 kL | 24. 3.25 h |
| 7. 0.4 cm | 16. 9 100 g | 25. 180 h |
| 8. 56 000 L | 17. 285 min | 26. 9.75 h |
| 9. 1.8 km | 18. 555 min | 27. 186 h |

28. 7.5 h

Extension

1. about 36 792 000
2. 11.6 days

Exit ticket

1. Level 1: 450 cm
2. Level 2: 2.75 kg
3. Level 3: 210 min

8.02 Perimeter (booklet p. 9)

WE DO examples (teacher)

1. Ex 1a: 52 m
2. Ex 1b: 36 mm
3. Ex 2a: 78 cm
4. Ex 2b: 650 cm
5. Ex 3a: 128 mm
6. Ex 3b: 11.5 cm

Easy • Set A

1. 46 cm
2. 68 m
3. 36 mm
4. 38 cm
5. 13.6 m
6. 30 cm

Easy • Set A2 (quick drill)

1. 28 cm
2. 22 m
3. 15 mm
4. 33 cm
5. 36 m
6. 10 km

Easy • Set B

1. 78 cm
2. 52 m
3. 128 mm

Medium • Set C

1. 136 m
2. 74 cm
3. 190 mm
4. 650 cm
5. 11.5 cm
6. 6 m

Challenging • Set D

1. \$2597.00
2. 9.5 cm
3. 5 (1×9, 2×8, 3×7, 4×6, 5×5)
4. 66 cm

Skill drill 1 (Easy basics)

- | | | |
|----------|----------|------------|
| 1. 34 m | 11. 52 m | 21. 38 m |
| 2. 96 m | 12. 52 m | 22. 72 m |
| 3. 10 m | 13. 16 m | 23. 62 m |
| 4. 14 m | 14. 92 m | 24. 36 m |
| 5. 60 m | 15. 48 m | 25. 40 cm |
| 6. 36 m | 16. 56 m | 26. 85 cm |
| 7. 74 m | 17. 60 m | 27. 192 cm |
| 8. 6 m | 18. 74 m | 28. 52 cm |
| 9. 72 m | 19. 62 m | 29. 96 cm |
| 10. 16 m | 20. 58 m | 30. 24 cm |

31. 66 cm	35. 9 cm	39. 88 cm	4. 59.7 cm	15. 119.4 cm	26. 50.3 m
32. 60 cm	36. 48 cm	40. 54 cm	5. 88.0 cm	16. 56.5 cm	27. 12.6 m
33. 128 cm	37. 102 cm		6. 18.8 cm	17. 84.8 cm	28. 56.5 m
34. 48 cm	38. 36 cm		7. 69.1 cm	18. 50.3 cm	29. 18.8 m
			8. 34.6 cm	19. 62.8 cm	30. 75.4 m
			9. 110.0 cm	20. 122.5 cm	31. 88.0 m
			10. 40.8 cm	21. 37.7 m	32. 44.0 m
			11. 31.4 cm	22. 125.7 m	33. 69.1 m
			12. 15.7 cm	23. 94.2 m	34. 6.3 m
			13. 75.4 cm	24. 100.5 m	35. 131.9 m
			14. 6.3 cm	25. 31.4 m	36. 81.7 m

Skill drill 2 (Medium basics)

1. 13 m	14. 5 m	21. 18.8 m (or 1880 cm)
2. 9 m	15. 22 m	22. 8.46 m (or 846 cm)
3. 4 m	16. 2.26 m (or 226 cm)	23. 18.36 m (or 1836 cm)
4. 3 m	17. 6.92 m (or 692 cm)	24. 17.28 m (or 1728 cm)
5. 3 m	18. 17.42 m (or 1742 cm)	25. 10.86 m (or 1086 cm)
6. 5 m	19. 12.24 m (or 1224 cm)	26. 5.7 m (or 570 cm)
7. 3 m	20. 11.64 m (or 1164 cm)	27. 18.4 m (or 1840 cm)
8. 4 m		
9. 5 m		
10. 2 m		
11. 10 m		
12. 23 m		
13. 14 m		

Extension

- 10 cm by 15 cm
- 14 cm

Exit ticket

- Level 1: 38 cm
- Level 2: 28.5 m
- Level 3: 15 mm

8.03 Circumference of a circle (booklet p. 17)

WE DO examples (teacher)

- Ex 1a: 6.3 cm
- Ex 1b: 62.8 mm
- Ex 2a: 44.0 cm
- Ex 2b: 175.9 mm
- Ex 3a: 207.3 m
- Ex 3b: 15.9 cm

Easy · Set A (quick drill)

1. 10 cm	5. 42 cm	9. 18.8 cm
2. 24 m	6. 18 m	10. 31.4 mm
3. 7 mm	7. 12.6 cm	11. 3.1 m
4. 1.6 km	8. 31.4 m	12. 125.7 cm

Easy · Set B

- 6.3 cm
- 44.0 cm
- 22.0 m
- 94.2 mm
- 4.7 km
- 314.2 cm

Medium · Set C

- 62.8 mm
- 175.9 mm
- 18.8 cm
- 28.3 m
- 78.5 cm
- 3.1 km

Challenging · Set D

- 207.3 m
- 15.9 cm
- 30.8 cm
- 12 (circumference 377.0 cm)

Skill drill 1 (Easy basics)

1. 100.5 cm	2. 44.0 cm	3. 28.3 cm
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Skill drill 2 (Medium basics)

1. 46.8 cm	9. 11.5 cm	17. 28.3 cm
2. 19.1 cm	10. 29.6 cm	18. 66.8 cm
3. 40.7 cm	11. 28.6 cm	19. 41.1 cm
4. 37.9 cm	12. 22.9 cm	20. 12.9 cm
5. 17.2 cm	13. 51.4 cm	21. 30.8 cm
6. 12.7 cm	14. 25.7 cm	22. 59.1 cm
7. 22.6 cm	15. 43.7 cm	23. 38.6 cm
8. 29.3 cm	16. 18.0 cm	24. 64.3 cm

Extension

- 36.0 m
- 401.1 m

Exit ticket

- Level 1: 16 cm
- Level 2: 28.3 m
- Level 3: 22.0 cm

8.04 Area units (booklet p. 25)

WE DO examples (teacher)

- Ex 1a: a 1 cm square is 10 mm by 10 mm
- Ex 1b: 91 000
- Ex 2a: 250
- Ex 2b: 17.5
- Ex 3a: 0.24
- Ex 3b: 8 600 000

Easy · Set A (quick drill)

1. 300 mm ²	5. 2 000 000 m ²	9. 30 000 cm ²
2. 250 mm ²	6. 50 000 m ²	10. 2 m ²
3. 91 000 cm ²	7. 200 mm ²	11. 40 000 m ²
4. 40 000 cm ²	8. 5 cm ²	12. 3 ha

Easy · Set B

- 2.4 cm²
- 17.5 m²
- 5 ha
- 8.5 cm²
- 3.5 km²
- 0.64 m²

Medium · Set C

- 0.24 m²
- 8 600 000 mm²
- 7 ha
- 125 000 m²
- 3500 cm²
- 0.0072 m²

Challenging · Set D

- mm² or cm²; m²; ha; km²
- 3.2 m² (30 000 cm² = 3 m²)

3. 150 tiles
4. 240 ha

Skill drill 1 (Easy basics)

- | | | |
|----------------------------|-----------------------------|-----------------------------|
| 1. 5 400 cm ² | 15. 60 000 cm ² | 29. 270 000 cm ² |
| 2. 49 km ² | 16. 4.6 cm ² | 30. 280 000 cm ² |
| 3. 4 300 mm ² | 17. 0.37 km ² | 31. 400 mm ² |
| 4. 2 900 ha | 18. 10 000 cm ² | 32. 100 000 cm ² |
| 5. 27 cm ² | 19. 52 ha | 33. 240 000 cm ² |
| 6. 3 300 mm ² | 20. 5.6 cm ² | 34. 170 000 cm ² |
| 7. 26 ha | 21. 3 000 mm ² | 35. 290 000 cm ² |
| 8. 12 000 cm ² | 22. 100 mm ² | 36. 1 100 mm ² |
| 9. 41 000 cm ² | 23. 1 cm ² | 37. 1 900 mm ² |
| 10. 3 100 mm ² | 24. 180 ha | 38. 190 000 cm ² |
| 11. 60 km ² | 25. 300 000 cm ² | 39. 600 mm ² |
| 12. 4 100 cm ² | 26. 10 000 cm ² | 40. 260 000 cm ² |
| 13. 4 500 mm ² | 27. 200 mm ² | |
| 14. 530 000 m ² | 28. 2 400 mm ² | |

Skill drill 2 (Medium basics)

- | | | |
|----------------------------|-----------------------------|---------------------------|
| 1. 400 mm ² | 11. 70 mm ² | 21. cm ² |
| 2. 7 600 ha | 12. 75 m ² | 22. m ² |
| 3. 790 mm ² | 13. 320 000 cm ² | 23. mm ² |
| 4. 62 000 cm ² | 14. 1 800 ha | 24. km ² or ha |
| 5. 530 mm ² | 15. 6 m ² | 25. ha |
| 6. 37 ha | 16. 6 600 ha | 26. cm ² |
| 7. 21 000 m ² | 17. m ² | 27. m ² or ha |
| 8. 60 000 cm ² | 18. km ² | 28. m ² |
| 9. 670 000 m ² | 19. km ² | |
| 10. 980 000 m ² | 20. cm ² | |

Extension

1. 1 000 000 mm²
2. 0.71 ha

Exit ticket

1. Level 1: 500 mm²
2. Level 2: 3 m²
3. Level 3: 1 500 000 mm²

8.05 Area of squares and rectangles

(booklet p. 31)

WE DO examples (teacher)

1. Ex 1a: 16 m²
2. Ex 1b: 24 cm²
3. Ex 2a: 162 m²
4. Ex 2b: 1.5 m²
5. Ex 3a: 9 cm
6. Ex 3b: 6 m

Easy · Set A

1. 16 m²
2. 24 cm²
3. 162 m²
4. 60 mm²
5. 49 cm²
6. 3 m²

Easy · Set A2 (quick drill)

1. 36 cm²
2. 28 m²
3. 60 mm²

4. 2.25 m²
5. 10 cm²
6. 121 km²

Easy · Set B

1. 120 cm²
2. 121 m²
3. 7 m²
4. 0.25 km²
5. 1000 mm²
6. 0.6 m²

Medium · Set C

1. 9 cm
2. 6 m
3. 3 cm
4. 1.2 m
5. 1.5 m²
6. 81 cm²

Challenging · Set D

1. \$936
2. 108 tiles
3. 8 cm by 5 cm
4. it is 4 times as large

Skill drill 1 (Easy basics)

- | | | |
|-----------------------|-------------------------|-------------------------|
| 1. 112 m ² | 15. 80 m ² | 29. 225 cm ² |
| 2. 4 m ² | 16. 34 m ² | 30. 64 cm ² |
| 3. 40 m ² | 17. 56 m ² | 31. 1 cm ² |
| 4. 68 m ² | 18. 8 m ² | 32. 81 cm ² |
| 5. 256 m ² | 19. 16 m ² | 33. 324 cm ² |
| 6. 119 m ² | 20. 30 m ² | 34. 196 cm ² |
| 7. 40 m ² | 21. 216 m ² | 35. 36 cm ² |
| 8. 70 m ² | 22. 99 m ² | 36. 16 cm ² |
| 9. 6 m ² | 23. 36 m ² | 37. 169 cm ² |
| 10. 36 m ² | 24. 100 m ² | 38. 289 cm ² |
| 11. 72 m ² | 25. 361 cm ² | 39. 25 cm ² |
| 12. 22 m ² | 26. 4 cm ² | 40. 9 cm ² |
| 13. 9 m ² | 27. 49 cm ² | |
| 14. 15 m ² | 28. 100 cm ² | |

Skill drill 2 (Medium basics)

- | | | |
|----------|--------------------------|--------------------------|
| 1. 8 m | 11. 3 m | 20. 11.4 cm ² |
| 2. 15 m | 12. 5 m | 21. 27 cm ² |
| 3. 16 m | 13. 15 m | 22. 35.4 cm ² |
| 4. 2 m | 14. 10 m | 23. 62 cm ² |
| 5. 6 m | 15. 9 m | 24. 40.5 cm ² |
| 6. 19 m | 16. 36 cm ² | 25. 64.8 cm ² |
| 7. 13 m | 17. 22.8 cm ² | 26. 89.1 cm ² |
| 8. 14 m | 18. 21.6 cm ² | 27. 45.5 cm ² |
| 9. 14 m | 19. 28.2 cm ² | |
| 10. 18 m | | |

Extension

1. 24 m²
2. 6 cm by 6 cm (36 cm²)

Exit ticket

1. Level 1: 54 cm²
2. Level 2: 6.25 m²
3. Level 3: 12 m

8.06 Area of a triangle (booklet p. 39)

WE DO examples (teacher)

1. Ex 1a: 360 mm²
2. Ex 1b: 112 mm²
3. Ex 2a: 5.2 cm²
4. Ex 2b: 187 km²
5. Ex 3a: 8 cm
6. Ex 3b: two copies of the triangle make a rectangle (or parallelogram)

Easy · Set A

1. 360 mm²
2. 30 cm²
3. 20 m²
4. 42 cm²
5. 5.2 cm²
6. 150 m²

Easy · Set A2 (quick drill)

1. 20 cm²
2. 21 m²
3. 30 mm²
4. 32 cm²
5. 7.5 m²
6. 90 km²

Easy · Set B

1. 12 cm²
2. 112 mm²
3. 54 m²
4. 17.5 cm²
5. 1.8 m²
6. 2500 cm²

Medium · Set C

1. 187 km²
2. 40 m²
3. 54 cm²
4. 8 cm
5. 5 m
6. 30 cm²

Challenging · Set D

1. 8 m²
2. 4 bags
3. 92 cm²
4. the second is twice the first (20 cm² vs 40 cm²)

Skill drill 1 (Easy basics)

- | | | |
|-------------------------|-------------------------|--------------------------|
| 1. 28 cm ² | 15. 66 cm ² | 29. 100 m ² |
| 2. 54 cm ² | 16. 85 cm ² | 30. 28 m ² |
| 3. 14 cm ² | 17. 48 cm ² | 31. 253 m ² |
| 4. 152 cm ² | 18. 52 cm ² | 32. 137.5 m ² |
| 5. 10 cm ² | 19. 36 cm ² | 33. 55 m ² |
| 6. 36 cm ² | 20. 17 cm ² | 34. 184 m ² |
| 7. 14 cm ² | 21. 51 cm ² | 35. 252 m ² |
| 8. 33 cm ² | 22. 54 cm ² | 36. 237.5 m ² |
| 9. 70 cm ² | 23. 38 cm ² | 37. 170 m ² |
| 10. 18 cm ² | 24. 20 cm ² | 38. 90 m ² |
| 11. 10 cm ² | 25. 152 m ² | 39. 237.5 m ² |
| 12. 150 cm ² | 26. 150 m ² | 40. 65 m ² |
| 13. 77 cm ² | 27. 49.5 m ² | |
| 14. 20 cm ² | 28. 198 m ² | |

Skill drill 2 (Medium basics)

- | | | |
|----------|----------|----------|
| 1. 8 cm | 3. 13 cm | 5. 8 cm |
| 2. 16 cm | 4. 3 cm | 6. 17 cm |

- | | | |
|-----------|-------------------------|-------------------------|
| 7. 20 cm | 13. 1.1 m ² | 19. 7.2 m ² |
| 8. 4 cm | 14. 26.8 m ² | 20. 17.6 m ² |
| 9. 11 cm | 15. 37 m ² | 21. 35.2 m ² |
| 10. 14 cm | 16. 32.8 m ² | 22. 2.8 m ² |
| 11. 4 cm | 17. 13.8 m ² | 23. 10.8 m ² |
| 12. 2 cm | 18. 5.2 m ² | 24. 12.9 m ² |

Extension

1. 27 m²
2. 24 square units

Exit ticket

1. Level 1: 35 m²
2. Level 2: 8 cm²
3. Level 3: 8 cm

8.07 Area of a parallelogram (booklet p. 47)

WE DO examples (teacher)

1. Ex 1a: 120 m²
2. Ex 1b: 828 cm²
3. Ex 2a: 68 mm²
4. Ex 2b: cutting off a triangle and moving it makes a rectangle
5. Ex 3a: 6 cm
6. Ex 3b: 50 cm²

Easy · Set A

1. 120 m²
2. 828 cm²
3. 68 mm²
4. 60 cm²
5. 180 m²
6. 31.5 cm²

Easy · Set A2 (quick drill)

1. 36 cm²
2. 120 m²
3. 13 mm²
4. 144 cm²
5. 45 m²
6. 77 km²

Easy · Set B

1. 60 cm²
2. 9 m²
3. 300 mm²
4. 28 km²
5. 0.96 m²
6. 450 cm²

Medium · Set C

1. 6 cm
2. 9 m
3. 2 m
4. 50 cm²
5. 98 mm²
6. parallelogram 48 cm², triangle 24 cm²

Challenging · Set D

1. \$882
2. e.g. 6 × 4 and 8 × 3
3. 80 cm²
4. 9 cm

Skill drill 1 (Easy basics)

- | | | |
|-----------------------|----------------------|----------------------|
| 1. 120 m ² | 2. 30 m ² | 3. 54 m ² |
|-----------------------|----------------------|----------------------|

- | | | |
|-----------------------|------------------------|------------------------|
| 4. 70 m^2 | 17. 90 m^2 | 30. 550 cm^2 |
| 5. 18 m^2 | 18. 171 m^2 | 31. 324 cm^2 |
| 6. 12 m^2 | 19. 104 m^2 | 32. 372 cm^2 |
| 7. 34 m^2 | 20. 153 m^2 | 33. 117 cm^2 |
| 8. 126 m^2 | 21. 168 m^2 | 34. 90 cm^2 |
| 9. 144 m^2 | 22. 24 m^2 | 35. 192 cm^2 |
| 10. 200 m^2 | 23. 45 m^2 | 36. 130 cm^2 |
| 11. 20 m^2 | 24. 80 m^2 | 37. 189 cm^2 |
| 12. 44 m^2 | 25. 132 cm^2 | 38. 180 cm^2 |
| 13. 88 m^2 | 26. 245 cm^2 | 39. 65 cm^2 |
| 14. 30 m^2 | 27. 348 cm^2 | 40. 459 cm^2 |
| 15. 60 m^2 | 28. 660 cm^2 | |
| 16. 9 m^2 | 29. 612 cm^2 | |

Skill drill 2 (Medium basics)

- | | | |
|---------|-------------------------|-------------------------|
| 1. 10 m | 10. 2 m | 18. 35.67 m^2 |
| 2. 12 m | 11. 4 m | 19. 34.79 m^2 |
| 3. 10 m | 12. 12 m | 20. 10.54 m^2 |
| 4. 6 m | 13. 13.44 m^2 | 21. 8.16 m^2 |
| 5. 8 m | 14. 11.75 m^2 | 22. 28 m^2 |
| 6. 17 m | 15. 14.56 m^2 | 23. 30.4 m^2 |
| 7. 14 m | 16. 15.18 m^2 | 24. 12.39 m^2 |
| 8. 5 m | 17. 5.22 m^2 | |
| 9. 6 m | | |

Extension

- 32 m^2
- 12 cm

Exit ticket

- Level 1: 36 cm^2
- Level 2: 4 m^2
- Level 3: 12.5 m

8.08 Area of composite shapes (booklet p. 55)

WE DO examples (teacher)

- Ex 1a: 23 cm^2
- Ex 1b: 71 m^2
- Ex 2a: 30 m^2
- Ex 2b: 360 m^2
- Ex 3a: 6 L
- Ex 3b: 37.5 mm^2

Easy • Set A

- 71 m^2
- 72 cm^2
- 105 m^2

Easy • Set A2 (quick drill)

- 56 m^2
- 23 cm^2
- 52 mm^2
- 30 m^2

Easy • Set B

- 23 cm^2
- 30 m^2
- 78 m^2

Medium • Set C

- 360 m^2
- 37.5 mm^2
- 272.4 mm^2
- 84 cm^2
- 80 m^2

- 55 cm^2

Challenging • Set D

- 6 L
- 630 cm^2
- 21 m^2
- \$798

Skill drill 1 (Easy basics)

- | | | |
|------------------------|------------------------|----------------------|
| 1. 96 cm^2 | 11. 99 cm^2 | 21. 74 m^2 |
| 2. 135 cm^2 | 12. 166 cm^2 | 22. 56 m^2 |
| 3. 101 cm^2 | 13. 147 cm^2 | 23. 94 m^2 |
| 4. 70 cm^2 | 14. 82 cm^2 | 24. 62 m^2 |
| 5. 170 cm^2 | 15. 66 m^2 | 25. 59 m^2 |
| 6. 111 cm^2 | 16. 51 m^2 | 26. 63 m^2 |
| 7. 60 cm^2 | 17. 90 m^2 | 27. 63 m^2 |
| 8. 116 cm^2 | 18. 38 m^2 | 28. 92 m^2 |
| 9. 177 cm^2 | 19. 126 m^2 | |
| 10. 157 cm^2 | 20. 20 m^2 | |

Skill drill 2 (Medium basics)

- | | | |
|--------------------------|--------------------------|-----------------------|
| 1. $60 (\text{units}^2)$ | 5. $48 (\text{units}^2)$ | 9. 99 m^2 |
| 2. $74 (\text{units}^2)$ | 6. $72 (\text{units}^2)$ | 10. 110 m^2 |
| 3. $68 (\text{units}^2)$ | 7. 30 m^2 | 11. 65 m^2 |
| 4. $58 (\text{units}^2)$ | 8. 68 m^2 | 12. 110 m^2 |

Extension

- 57 m^2
- 45 cm^2

Exit ticket

- Level 1: 46 cm^2
- Level 2: 22 m^2
- Level 3: 3 L

8.09 Prisms, cross-sections and views

(booklet p. 63)

WE DO examples (teacher)

- Ex 1a: yes; rectangle
- Ex 1b: yes; triangle
- Ex 2a: L-shape
- Ex 2b: no: the cross-section changes and the ends are not the same
- Ex 3a: front: L-shape; left: rectangle; top: rectangle
- Ex 3b: right: rectangle; front: pentagon (house shape); top: rectangle with a line

Easy • Set A

- yes, rectangle
- yes, triangle
- yes, trapezium
- yes, hexagon
- no
- not a prism (curved), but it has a circular cross-section

Easy • Set A2 (quick drill)

- square
- triangle
- hexagon
- rectangle
- circle
- pentagon

Easy • Set B

- L-shape
- pentagon

3. T-shape

Medium · Set C

1. L-shape; rectangle; rectangle
2. rectangle; pentagon; rectangle split by a line

Challenging · Set D

1. T-shape; rectangle; rectangle
2. a cylinder
3. a cube (or sphere)
4. a rectangle

Skill drill 1 (Easy basics)

- | | | |
|--------------|----------------------|---------------------|
| 1. circle | 8. rectangle | 15. F 6, V 8, E 12 |
| 2. hexagon | 9. pentagon | 16. F 7, V 10, E 15 |
| 3. hexagon | 10. triangle | 17. F 5, V 5, E 8 |
| 4. octagon | 11. square | 18. F 6, V 8, E 12 |
| 5. circle | 12. triangle | 19. F 4, V 4, E 6 |
| 6. rectangle | 13. F 10, V 16, E 24 | 20. F 5, V 6, E 9 |
| 7. square | 14. F 8, V 12, E 18 | |

Skill drill 2 (Medium basics)

- | | | |
|------------------------|-------------------|----------------------|
| 1. yes | 7. yes | 14. F 14, V 24, E 36 |
| 2. no | 8. yes | 15. F 9, V 14, E 21 |
| 3. yes | 9. yes | 16. F 8, V 12, E 18 |
| 4. no (curved surface) | 10. no | 17. F 6, V 8, E 12 |
| 5. no | 11. yes | 18. F 7, V 10, E 15 |
| 6. no | 12. yes | 19. F 12, V 20, E 30 |
| | 13. F 5, V 6, E 9 | 20. F 10, V 16, E 24 |

Extension

1. a square pyramid
2. three squares in a row of heights 1, 2, 3 (or one tall column, depending on view)

Exit ticket

1. Level 1: No: it has curved sides (it is like a prism with a circle cross-section)
2. Level 2: hexagon
3. Level 3: a rectangle with a line along the middle

8.10 Volume units (booklet p. 71)

WE DO examples (teacher)

1. Ex 1a: a 1 cm cube is $10 \times 10 \times 10$ mm
2. Ex 1b: 20 000
3. Ex 2a: 500 000
4. Ex 2b: 7.5
5. Ex 3a: 0.00023
6. Ex 3b: 60 cubes

Easy · Set A (quick drill)

- | | | |
|------------------------------|------------------------------|-------------------------------|
| 1. 20 000 mm ³ | 5. 4500 mm ³ | 9. 1 000 000 cm ³ |
| 2. 3000 mm ³ | 6. 1 200 000 cm ³ | 10. 0.5 m ³ |
| 3. 2 000 000 cm ³ | 7. 4000 mm ³ | 11. 300 mm ³ |
| 4. 500 000 cm ³ | 8. 2 cm ³ | 12. 2 500 000 cm ³ |

Easy · Set B

1. 7.5 cm³
2. 4 cm³
3. 3 m³
4. 0.25 m³
5. 0.6 cm³
6. 0.08 m³

Medium · Set C

1. 0.00023 m³
2. 4 000 000 mm³

3. 5 000 000 000 m³

4. 750 000 cm³

5. 0.0125 m³

6. 0.000009 m³

Challenging · Set D

1. 60
2. 30
3. 15 000 cm³, 0.02 m³, 25 000 000 mm³
4. cm³; m³; cm³

Skill drill 1 (Easy basics)

- | | | |
|-------------------------------|-------------------------------|-------------------------|
| 1. 27 000 mm ³ | 14. 0.51 m ³ | 27. 7.33 L |
| 2. 38 000 000 cm ³ | 15. 60 mm ³ | 28. 1.32 L |
| 3. 0.16 cm ³ | 16. 60 000 mm ³ | 29. 368 mL |
| 4. 2.3 m ³ | 17. 4 000 mm ³ | 30. 8.48 L |
| 5. 9 000 mm ³ | 18. 290 000 cm ³ | 31. 334 mL |
| 6. 0.42 m ³ | 19. 4 200 000 cm ³ | 32. 538 cm ³ |
| 7. 14 cm ³ | 20. 33 m ³ | 33. 9.5 L |
| 8. 37 000 mm ³ | 21. 0.35 m ³ | 34. 642 mL |
| 9. 53 000 000 cm ³ | 22. 56 m ³ | 35. 8.14 L |
| 10. 3 cm ³ | 23. 0.44 m ³ | 36. 0.62 L |
| 11. 540 000 cm ³ | 24. 4 600 mm ³ | 37. 4.38 L |
| 12. 6 cm ³ | 25. 640 cm ³ | 38. 559 cm ³ |
| 13. 600 mm ³ | 26. 466 mL | 39. 0.95 L |
| | | 40. 6.49 L |

Skill drill 2 (Medium basics)

- | | | |
|------------------------------|-------------------------------|------------------------|
| 1. 7 600 mm ³ | 10. 4 100 000 cm ³ | 20. 15 m ³ |
| 2. 3 800 mm ³ | 11. 8 900 mm ³ | 21. 47 m ³ |
| 3. 8 000 mm ³ | 12. 46 cm ³ | 22. 77 m ³ |
| 4. 4.7 m ³ | 13. 4 500 000 cm ³ | 23. 30 m ³ |
| 5. 7.7 cm ³ | 14. 4.9 cm ³ | 24. 53 000 L |
| 6. 0.2 m ³ | 15. 59 m ³ | 25. 32 m ³ |
| 7. 8.7 cm ³ | 16. 2.8 cm ³ | 26. 0.1 m ³ |
| 8. 5 200 000 cm ³ | 17. 43 000 L | 27. 3.1 m ³ |
| 9. 31 cm ³ | 18. 61 000 L | 28. 7.1 m ³ |
| | 19. 3 000 L | |

Extension

1. 350 000 mm³
2. 1000

Exit ticket

1. Level 1: 6000 mm³
2. Level 2: 2.5 m³
3. Level 3: 1.5 cm³

8.11 Volume of a rectangular prism (booklet p. 77)

WE DO examples (teacher)

1. Ex 1a: 80 m³
2. Ex 1b: 320 cm³
3. Ex 2a: 672 mm³
4. Ex 2b: 3000 cm³
5. Ex 3a: 4 cm
6. Ex 3b: 5 cm

Easy · Set A

1. 80 m³
2. 320 cm³
3. 672 mm³
4. 360 m³

5. 3000 cm³
6. 27 cm³

Easy · Set A2 (quick drill)

1. 40 cm³
2. 27 m³
3. 300 mm³
4. 24 m³
5. 1000 cm³
6. 108 cm³

Easy · Set B

1. 60 cm³
2. 125 m³
3. 5 m³
4. 300 mm³
5. 120 cm³
6. 0.3 m³

Medium · Set C

1. 4 cm
2. 5 cm
3. 6 m
4. 10 mm
5. 0.03 m³
6. 8 times

Challenging · Set D

1. 74.88 m³
2. 72 cubes
3. 0.96 m³
4. e.g. 1×1×48, 2×4×6, 3×4×4

Skill drill 1 (Easy basics)

- | | | |
|-------------------------|--------------------------|--------------------------|
| 1. 60 cm ³ | 15. 112 cm ³ | 29. 1331 cm ³ |
| 2. 288 cm ³ | 16. 32 cm ³ | 30. 125 m ³ |
| 3. 198 cm ³ | 17. 315 cm ³ | 31. 343 cm ³ |
| 4. 108 cm ³ | 18. 144 cm ³ | 32. 1000 m ³ |
| 5. 108 cm ³ | 19. 144 cm ³ | 33. 125 cm ³ |
| 6. 630 cm ³ | 20. 462 cm ³ | 34. 216 m ³ |
| 7. 162 cm ³ | 21. 18 cm ³ | 35. 8 cm ³ |
| 8. 252 cm ³ | 22. 336 cm ³ | 36. 512 m ³ |
| 9. 32 cm ³ | 23. 400 cm ³ | 37. 1000 cm ³ |
| 10. 96 cm ³ | 24. 140 cm ³ | 38. 64 cm ³ |
| 11. 192 cm ³ | 25. 1728 cm ³ | 39. 8 m ³ |
| 12. 63 cm ³ | 26. 1 cm ³ | 40. 512 cm ³ |
| 13. 210 cm ³ | 27. 216 cm ³ | |
| 14. 90 cm ³ | 28. 729 m ³ | |

Skill drill 2 (Medium basics)

- | | | |
|---------|------------------------|--------------------------|
| 1. 5 cm | 10. 7 cm | 18. 51.6 m ³ |
| 2. 6 cm | 11. 7 cm | 19. 116 m ³ |
| 3. 3 cm | 12. 2 cm | 20. 114 m ³ |
| 4. 7 cm | 13. 168 m ³ | 21. 64.8 m ³ |
| 5. 7 cm | 14. 55 m ³ | 22. 38.4 m ³ |
| 6. 8 cm | 15. 35 m ³ | 23. 156.8 m ³ |
| 7. 4 cm | 16. 106 m ³ | 24. 65 m ³ |
| 8. 6 cm | 17. 32 m ³ | |
| 9. 9 cm | | |

Extension

1. 540 cm³
2. 108 s

Exit ticket

1. Level 1: 60 cm³
2. Level 2: 216 m³
3. Level 3: 6 cm

8.12 Volume of a prism (booklet p. 85)

WE DO examples (teacher)

1. Ex 1a: 554.472 cm³
2. Ex 1b: 2200 mm³
3. Ex 2a: 570 m³
4. Ex 2b: 280 cm³
5. Ex 3a: 120 cm³
6. Ex 3b: 30 m²

Easy · Set A (quick drill)

1. 100 cm³
2. 280 cm³
3. 36 m³
4. 30 mm³
5. 50 cm³
6. 480 m³
7. 60 cm³
8. 63 m³
9. 25 cm³
10. 120 mm³

Easy · Set B

1. 120 cm³
2. 2200 mm³
3. 144 m³

Medium · Set C

1. 554.472 cm³
2. 570 m³
3. 540 cm³

Challenging · Set D

1. 5.4 m³
2. 0.6 m³
3. 30 m²
4. 300 000 cm³

Skill drill 1 (Easy basics)

- | | | |
|-------------------------|-------------------------|-------------------------|
| 1. 696 cm ³ | 13. 148 cm ³ | 25. 324 cm ³ |
| 2. 40 cm ³ | 14. 234 cm ³ | 26. 462 cm ³ |
| 3. 33 cm ³ | 15. 290 cm ³ | 27. 9 cm ³ |
| 4. 440 cm ³ | 16. 318 cm ³ | 28. 420 cm ³ |
| 5. 286 cm ³ | 17. 308 cm ³ | 29. 450 cm ³ |
| 6. 56 cm ³ | 18. 81 cm ³ | 30. 216 cm ³ |
| 7. 84 cm ³ | 19. 462 cm ³ | 31. 210 cm ³ |
| 8. 171 cm ³ | 20. 212 cm ³ | 32. 180 cm ³ |
| 9. 153 cm ³ | 21. 330 cm ³ | 33. 840 cm ³ |
| 10. 270 cm ³ | 22. 280 cm ³ | 34. 126 cm ³ |
| 11. 56 cm ³ | 23. 116 cm ³ | 35. 48 cm ³ |
| 12. 468 cm ³ | 24. 546 cm ³ | 36. 168 cm ³ |

Skill drill 2 (Medium basics)

- | | | |
|---------|------------------------|--------------------------|
| 1. 8 m | 9. 9 m | 16. 137.5 m ³ |
| 2. 14 m | 10. 14 m | 17. 126 m ³ |
| 3. 9 m | 11. 4 m | 18. 56 m ³ |
| 4. 7 m | 12. 2 m | 19. 810 m ³ |
| 5. 4 m | 13. 360 m ³ | 20. 595 m ³ |
| 6. 5 m | 14. 136 m ³ | 21. 544 m ³ |
| 7. 13 m | 15. 264 m ³ | |
| 8. 3 m | | |

Extension

1. 1600 m³
2. 375 000 L

Exit ticket

1. Level 1: 90 cm³
2. Level 2: 90 m³
3. Level 3: 8 cm

8.13 Capacity (booklet p. 93)

WE DO examples (teacher)

1. Ex 1a: 6 L
2. Ex 1b: 15 cm (D)
3. Ex 2a: 3500 L
4. Ex 2b: 80 000 L
5. Ex 3a: 2 000 000 000 ML
6. Ex 3b: 8 cups

Easy · Set A (quick drill)

- | | | |
|-----------|----------------|------------|
| 1. 500 mL | 5. 4.5 kL | 9. 2000 L |
| 2. 2 L | 6. 2 000 000 L | 10. 3.5 L |
| 3. 3500 L | 7. 1 L | 11. 6000 L |
| 4. 200 L | 8. 250 mL | 12. 500 L |

Easy · Set B

1. 6 L
2. 80 000 L
3. 0.32 L
4. 60 L
5. 1000 L
6. 1 L

Medium · Set C

1. 0.672 mL → 0.000672 L
2. 360 000 L
3. 3 L
4. 0.0022 L
5. 570 000 L
6. 15 625 L

Challenging · Set D

1. 15 cm
2. 2 000 000 000 ML
3. 63 L
4. 10 cm

Skill drill 1 (Easy basics)

- | | | |
|--------------|--------------------------|--------------------------|
| 1. 0.05 kL | 15. 1 L | 28. 1200 cm ³ |
| 2. 1.6 L | 16. 5.6 L | 29. 3.5 L |
| 3. 5 000 L | 17. 0.39 L | 30. 3800 cm ³ |
| 4. 19 000 mL | 18. 1 700 mL | 31. 4800 cm ³ |
| 5. 58 000 L | 19. 53 000 L | 32. 1000 cm ³ |
| 6. 300 mL | 20. 3 600 mL | 33. 1700 cm ³ |
| 7. 57 000 L | 21. 510 L | 34. 2800 cm ³ |
| 8. 580 mL | 22. 47 kL | 35. 3.2 L |
| 9. 420 mL | 23. 0.16 kL | 36. 4.1 L |
| 10. 3 900 L | 24. 3.5 L | 37. 1.8 L |
| 11. 17 L | 25. 4300 cm ³ | 38. 4400 cm ³ |
| 12. 18 L | 26. 800 cm ³ | 39. 3.1 L |
| 13. 130 mL | 27. 3.9 L | 40. 2300 cm ³ |
| 14. 21 L | | |

Skill drill 2 (Medium basics)

- | | | |
|-----------|-------------|-------------|
| 1. 10 L | 9. 12 L | 17. 10 cups |
| 2. 12.5 L | 10. 3 L | 18. 30 cups |
| 3. 62.5 L | 11. 25 L | 19. 6 cups |
| 4. 12 L | 12. 20 L | 20. 12 cups |
| 5. 60 L | 13. 18 cups | 21. 30 cups |
| 6. 16 L | 14. 24 cups | 22. 20 cups |
| 7. 9 L | 15. 36 cups | 23. 10 cups |
| 8. 24 L | 16. 45 cups | 24. 4 cups |

Extension

1. 50 mm
2. 1½ days

Exit ticket

1. Level 1: 2.5 L
2. Level 2: 5 L
3. Level 3: 600 L